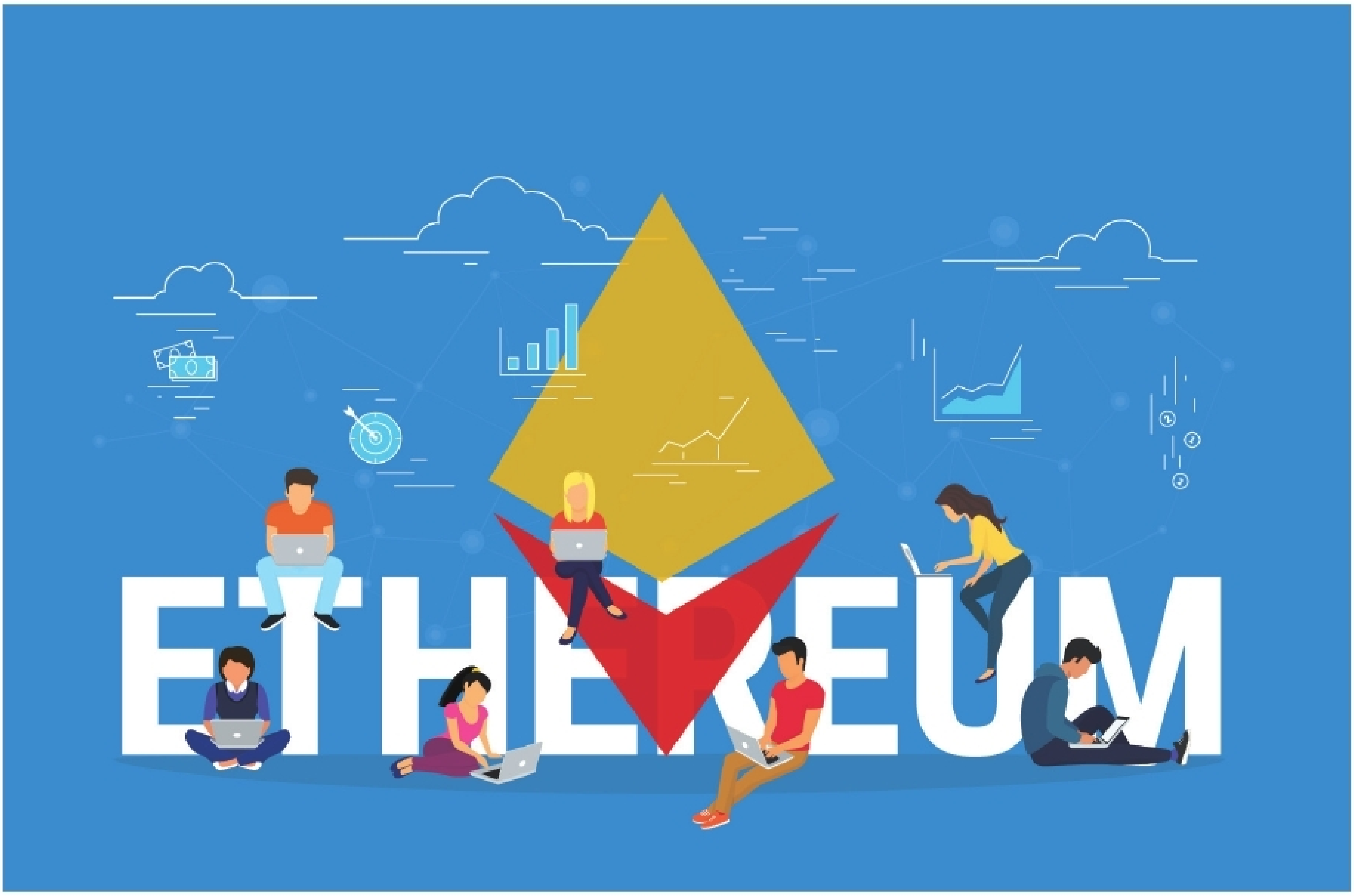
Discover Ethereum, a Blockchain Based Computing Platform

Dive into the world of blockchains, and use Ethereum to host a private network and deploy a smart contract.



Blockchain, as you might already know, is a tamper-resistant, append-only, distributed ledger. It is expected to be the next disruptive technology, although some argue that it is more of a foundational technology than something disruptive. Bitcoin was the first large scale adoption of a blockchain based platform, but the blockchain technology is capable of supporting much more than just cryptocurrencies. Ethereum provides an open source blockchain based distributed computing platform on which we can deploy apps to do much more than just financial transactions or money exchange. I've chosen Ethereum because it is very popular, and currently ranks second behind bitcoin, in terms of market capitalisation. Besides, it is open source, provides a Turing complete virtual machine – the Ethereum Virtual Machine (EVM) — to deploy the applications, and has an active community to support it.

In this article, we will discuss how to set up a private network that will host the Ethereum platform, including the blockchain. We will also learn to deploy a very basic application and interact with it. First, let us discuss some key terms that will help us understand the later sections.

Smart contract: This is a piece of code that is deployed in a distributed manner across the nodes hosting the network. It administers laws and rules, determines penalties and enforces them across the network. It contains states that are stored on the blockchain.

Consensus: This is the process by which all the different nodes agree about a state change. It ensures that even if malicious parties behave arbitrarily, the honest parties would still be able to reach an agreement.

Transactions: Any change of state is proposed as a transaction. The transactions (individually or collectively) are consented upon and then written into the blockchain.

Queries: Reading states from a current blockchain is known as a query. It does not require consensus, and any node can perform a query by just reading the latest states on the blockchain.